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Palaeomagnetism of some late Mesozoic dolerite dykes of South Spitsbergen

S. A. Vincenz, D. Cossack* and S. J. Duda† *Department of Earth and Atmospheric Sciences, Saint Louis University, St Louis, Missouri 63156, USA*

K. Birkenmajer *Institute of Geology, Polish Academy of Sciences, Cracow, Poland*

M. Jeleńska, M. Kądziałko-Hofmokl and J. Kruczyk *Institute of Geophysics, Polish Academy of Sciences, Warsaw, Poland*

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Summary. Opaque mineralogical studies of late Mesozoic dolerite dykes of Wedel-Jarlsberg Land, South Spitsbergen, show that the dykes have undergone a mild hydrothermal alteration. The alteration probably took place at the time of consolidation of the dykes, but may also have occurred as a side effect of Tertiary orogeny, which affected the western parts of Spitsbergen. Seven of these dykes were sampled during two summer seasons, when samples were collected from different exposures. A conventional procedure based on the minimum scatter principle and study of Zijderveld diagrams applied to 282 specimens demagnetized by alternating fields and thermally reveals several 'characteristic remanence' directions. In three dykes the directions are consistently reversed, but one carries both normal and reversed remanence. The reversed directions can be attributed to a geomagnetic field reversal in the late Mesozoic and K-Ar dating of one of the reversed dykes gives an age of 110 ± 5 Myr BP in agreement with the known age of early Cretaceous field reversal.

Using a least squares method of analysis of step demagnetization, low coercivity components, not separated by the conventional procedure, have been successfully isolated in two of the dykes. The analysis applied to all the other dykes confirms the results of the conventional method, yielding comparable mean directions of magnetization. The conventional method applied to 'characteristic' directions from five dykes gives a mean palaeomagnetic pole at 210.1°, 78.9° N, $\alpha_{95} = 3.5°$ and the least squares procedure applied to directions from six dykes yields a pole at 222.7°, 69.7° N, $\alpha_{95} = 7.9°$. The pole positions obtained agree with those obtained for Spitsbergen dolerites by

*Deceased.

†Present address: Institut für Geophysik, Universität Hamburg, Hamburg, Federal Republic of Germany.

other workers. A mean late Mesozoic pole, computed from four available Spitsbergen poles, is east of the North American and west of the Eurasian Cretaceous poles, suggesting that the rupture involving North America, Spitsbergen, and Eurasia must have occurred as a post-Cretaceous event.

1 Introduction

Svalbard Archipelago of which Spitsbergen is the main island, lies on the edge of the Eurasian tectonic plate close to the De Geer/Nansen fracture zone. Since the Archipelago is situated within 300 km of the northern end of the Mid-Atlantic Ridge, it must have played an active role in the opening of the Arctic Ocean (Fig. 1a). The exact configuration of tectonic plate boundaries in the Arctic Basin is not known and information on past plate movements in this region would be of great interest. The results described below constitute the initial stage of a wider palaeomagnetic project involving rocks of various ages exposed on Svalbard (Jeleńska, Kądziałko-Hofmokl & Kruczyk 1978a; Jeleńska *et al.* 1978b; Jeleńska *et al.* 1978/79; Vincenz, Jeleńska & Aiinehsazian 1978; Vincenz *et al.* 1981).

So far, most of the palaeomagnetic investigations of Spitsbergen have concentrated on rocks in the western and north-western parts of the island and the information derived from these studies has been ambiguous (Krumsiek, Nagel & Nairn 1966; Spall 1968; Storetvedt 1972; Halvorsen 1972, 1973).

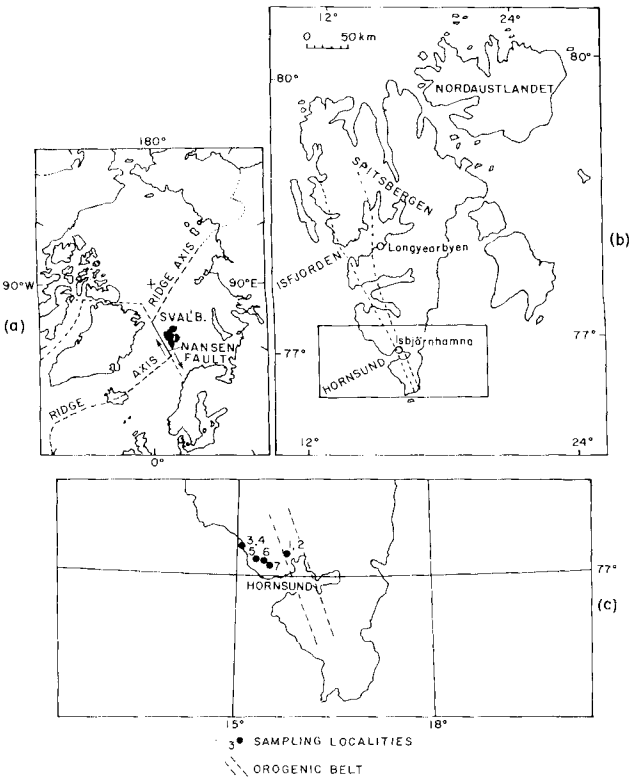


Figure 1. (a) Location of Svalbard in relation to Greenland, Scandinavia and Arctic/Atlantic spreading centres. (b) Sketch map of Spitsbergen, and (c) locations of sampling sites. The boundaries of the orogenic belt are approximate and indicate merely the zone of maximum tectonic disturbance.

Sandal & Halvorsen (1973) and Briseid & Halvorsen (1974) studied the magnetization of the late Mesozoic igneous rocks of north-east Spitsbergen and found that some units have apparently preserved the original cooling thermoremanence and give reliable palaeomagnetic pole positions.

The ambiguity of previous palaeomagnetic data has been ascribed by the above authors to the effects of West Spitsbergen orogeny. This orogeny affected the western edge of Spitsbergen during early to mid-Tertiary and was associated with the dextral displacement of Greenland (Harland 1969; Birkenmajer 1972). The tectonic effects of the orogeny are discernible within a wide north-south trending belt (Fig. 1b) embracing most of the Isfjorden region (Birkenmajer 1972). The orogenic belt is relatively narrow in the south of the island (Fig. 1c) where the effects of the orogeny are expected to be less pronounced. A suite of dolerite dykes crops out in this region in the area of Wedel-Jarlsberg Land north of Hornsund and seven of these dykes were selected for study. The dykes, generally assumed to be of late Mesozoic date, are accessible from the Polish research base at Isbjørnhamna on the north coast of Hornsund and use of this base facilitated the logistics of field operations.

2 Geology and mineralogy

The dolerite dykes of Wedel-Jarlsberg Land have been described by Birkenmajer & Morawski (1960). The dykes are independent of the Caledonian structure of the region and extend discordantly across anticlines and synclines. They have been intruded into the Precambrian sediments, metamorphics and granites, are vertical or near vertical and strike in various directions.

Birkenmajer & Morawski ascribed late Mesozoic age to these dykes. Using the petrographical classification of Spitsbergen dolerites (diabases, according to American terminology) due to Tyrell & Sandford (1933), the dykes belong to class A of the four classes, which comprises medium-to-fine-grained rocks typical for the majority of the Spitsbergen dolerites occurring both in the north-east and west of the island. In A-type dolerites titanomagnetite (TM) and ilmenite are usually found as skeletal aggregates often intergrown with pyroxenes, and here and there accessory pyrite.

The dykes have been subjected to alteration typified by sericitization of plagioclases, a process which usually occurs at relatively low temperatures (Goettel 1977, private communication). It is not clear whether the alteration was associated with the differentiation of the dykes, or was also caused by the West Spitsbergen orogeny. The sericites do not display any preferential orientation, suggesting a static and metasomatic character of the alteration process in a tectonically quiet environment. This, and the fact that the dykes are vertical and strike in several directions, suggests that they have not been seriously disturbed from their original positions, even within the orogenic belt. An almost complete absence of epidote in the dykes indicates that the alteration, whether contemporaneous or post-formational, did, in fact, proceed at relatively low temperatures and certainly below 300°C (Ade-Hall, Palmer & Hubbard 1971).

The general location of the sampling sites is shown in Fig. 1 which shows that five of the seven dykes lie outside the main orogenic belt. The boundaries of the belt shown are approximate and merely indicate the area of maximum tectonic disturbance.

Sampling was conducted during two summer seasons. The first collection was made in 1974 when most of the samples were obtained in the form of oriented cores drilled *in situ*, though in several cases oriented larger block samples were also taken. Orientation was performed using a Brunton compass or equivalent. Special care was taken to check for the effect of the magnetization of rock face on the magnetic needle, but fortunately no such effect was observed at any of the sampling sites. On average two or three cylindrical speci-

mens (2.49 cm diameter and 2.28 cm long) were derived from a core drilled *in situ* or in a block sample. Measurements were made at Saint Louis University, using a Princeton Applied Research, Inc. (PAR) Model SM-1 spinner magnetometer. The second collection was conducted in 1975 and involved only oriented block samples which were in most cases obtained from exposures other than those cored a year earlier. The measurable specimens took the form of 2 cm cubes and measurements were made at the Institute of Geophysics, Polish Academy of Sciences in Warsaw using a J4 spinner magnetometer manufactured by Geofyzika n.p. of Brno, Czechoslovakia. Geological field studies showed that, in distinction to rather fresh dykes nos 3 and 4, dykes nos 1, 2, 5, 6 and 7 were weathered, but the degree of weathering varied from exposure to exposure and there were no indications of a multiple structure, each dyke constituting a single body probably corresponding to a single intrusive event. Specifically, dykes nos 1, 5 and 7 were fairly large intrusive bodies, quite extensive in their exposures at times 100 m wide.

Potassium-argon (K-Ar) dating was carried on carefully selected fresh dolerite material from the reversely magnetized dyke no. 6. Three analyses were made, giving a consistent result of 110 ± 5 Myr BP. An analysis, conducted on material from normally magnetized dyke no. 4 gave an age of 66.8 ± 4.3 Myr BP.

Scatter in the K-Ar age determinations has been ascribed to various causes. Musset *et al.* (1972) and Purdy *et al.* (1972) considered the scatter in ages of dykes and lavas in terms of excess or loss of radiogenic argon. In the light of the above quoted age determinations, presence of excess radiogenic argon appears unlikely, but its loss could have occurred following the low temperature oxidation process associated with regional hydrothermal alteration. While the younger age probably reflects loss of radiogenic argon, the observed difference in ages may in part also reflect a time interval between the individual intrusive events. The older age places the first intrusive events at the close of Aptian and the beginning of Albian.

3 Opaque mineralogy

Polished sections of representative samples from five dykes were investigated using an ore microscope. The dolerites reveal little or no effects of deuteric (high temperature) oxidation. The deuteric oxidation class of TM as defined by Wilson & Watkins (1967), Wilson, Haggerty & Watkins (1968) and Ade-Hall *et al.* (1968) is 1 or 2. The primary minerals are TM and ilmenite which are often in close association, but the latter is seldom a product of deuteric oxidation and is usually quite fresh and often accompanied by accessory pyrite. Grain sizes vary from 1 to 500 μm, with smaller sizes more frequent on the chilled margins of the dykes. Generally, larger grains (200–500 μm) are predominant, except in dykes nos 3 and 4 which contain finer magnetic material in the range 1–100 μm, with 5–20 μm being the most common. Most of the TM grains reveal effects of regional hydrothermal alteration (Ade-Hall *et al.* 1971) superimposed over the mild (if any) deuteric changes and characterized by various stages of granulation with little or no alteration to titanomaghemite and to titanohematite and similarly very rare alteration of ilmenite lamellae (if present) or of primary ilmenite. The overall picture is one of a low deuteric oxidation class material which has undergone regional hydrothermal alteration at temperatures up to 250°C (Ade-Hall *et al.* 1971, 1972). These characteristics are common to all rock material, irrespective of whether it was derived from within the main orogenic belt or from outside it, suggesting that the dykes have been equally affected by the hydrothermal process which, therefore, was unlikely to have been generated solely by the Tertiary orogeny. The above described characteristics are closely reminiscent of the observations made by Ade-Hall *et al.* (1972), Wilson *et al.* (1974) and Hall, Wilson & Dagley (1977) on Tertiary dykes of the western Isles of Scotland.

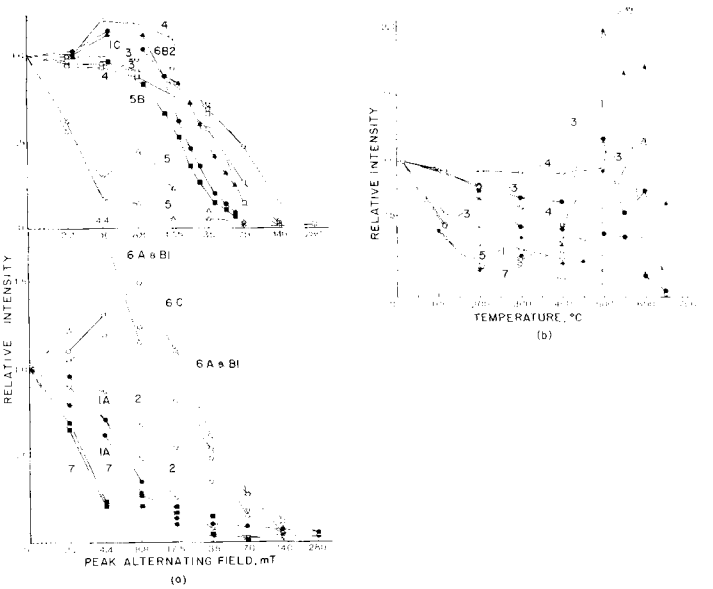


Figure 2. (a) Alternating field demagnetization curves of representative specimens. The designation of the specimens is the same as in Table 1, but where the behaviour of a specimen is typical of both sampling seasons, no capital letter designation is employed. (b) Thermal demagnetization curves of representative specimens. Specimen designation as in (a).

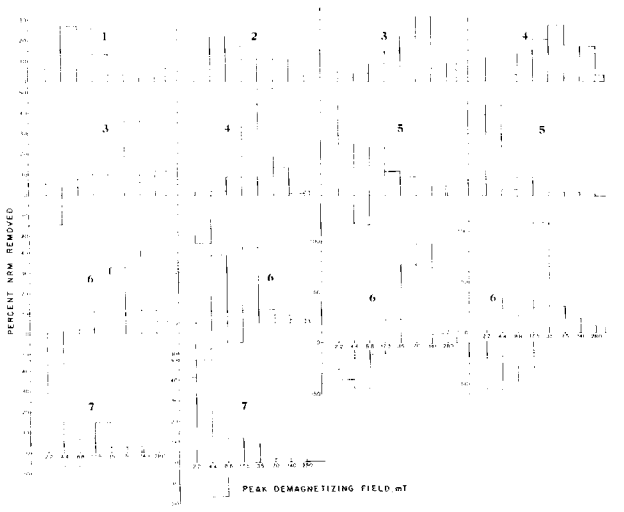


Figure 3. NRM coercivity spectra expressed as percentages of NRM removed between the given peak demagnetizing fields. Large numbers refer to numeration of the dykes.

4 Experimental results

For the most part, alternating field (AF) demagnetization was used to remove low coercivity components of natural remanence (NRM), but thermal demagnetization was also employed. Fig. 2 shows some typical AF and thermal demagnetization curves and Fig. 3 the corresponding NRM coercivity spectra expressed as percentages of NRM removed between the given peak demagnetizing fields used. Fig. 4 shows typical Zijderveld (1967) diagrams for



Figure 4. Zijderveld diagrams for representative specimens from each dyke. Large numbers refer to numeration of the dykes. The coordinates are those associated with each specimen investigated. Units: 10^{-1} Am^{-1}. Peak alternating fields in mT.

each dyke and Fig. 5 gives examples of directional changes on AF and thermal demagnetization of representative specimens from each dyke.

The highest stability of natural remanence occurs in dykes nos 3 and 4, while the remaining five dykes are far less stable, with the lowest stability occurring in dykes nos 5 and 7 (Fig. 2). AF demagnetizing curves of the material from dykes 1, 2, 3, 4 and 6 reveal NRM coercivities distributed over the whole peak demagnetizing field range from 0 to 280 mT (Fig. 3). In dykes 5 and 7 the coercivities tend to concentrate near the lower peak field values. In all dykes except in no. 2 reversed components of NRM are present. The coercivity spectra give a clear indication of the complex nature of the carriers of NRM in the dykes.

Thermal demagnetization curves (Fig. 2) show in most cases an unusual increase of remanence at temperatures exceeding 400°C (in rare cases, 300°C). This can be ascribed to either a phase change of widely present titanomaghemite or TM to maghemite or magnetite respectively (Jeleńska *et al.* 1978/79) or to generation of these minerals from other iron minerals (Vincenz *et al.* 1981). This factor does not render thermal demagnetization a desirable procedure for palaeomagnetic cleaning of Spitsbergen dolerites and although some specimens were thermally 'cleaned' by heating them up to 400°C, AF 'cleaning' was preferred.

The representative Zijderveld diagrams shown in Fig. 4 are thus based on AF demagnetization data only, since thermal demagnetization could not produce reliable results above 400°C. They show in all cases the presence of at least two components of NRM and more

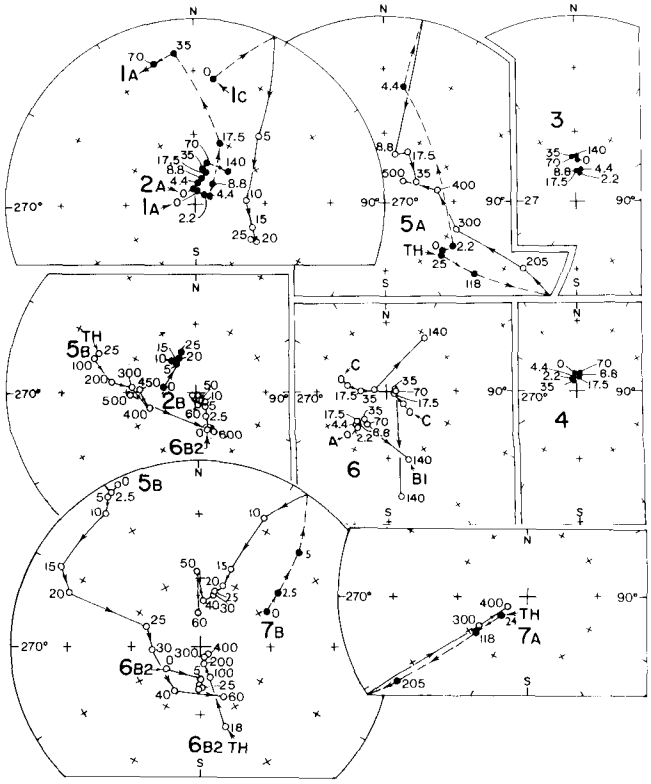


Figure 5. (a) Directional changes on AF and thermal (TH) demagnetization shown on equal area net: full circles, lower hemisphere; open circles, upper hemisphere. Large figures give dyke numbers and capitals specimen groups as defined in Table 1. Smaller figures give peak demagnetizing fields or heating temperatures.

often of three or even four components (dykes 1, 2, 5, 6 and 7). In several cases the final component is seen to decrease linearly towards the origin (dykes 1, 2, 3, 4 and 6, Fig. 4b, d, e, f and h), but unfortunately this decrease is not always clear or convincing (Fig. 4c, e and i).

The directional changes such as illustrated in Fig. 5 and in Zijderveld diagrams of Fig. 4 were used in attaining the best final direction. Using such information the most tightly grouped sets of points were sought on equal area net for each set of data, commensurate with the minimum spread in peak fields or heating temperatures and the beginning of linear decrease towards the origin on the Zijderveld plot. It was found that, in line with the information provided by Fig. 3, the demagnetization peak fields ranging from 8.8 to 35 mT and temperatures of 300°–400°C gave the best results. Table 1 gives the mean directions after such an analysis for the various groups of specimens from both collections.

The directions shown in Table 1 can be rearranged and divided into four distinct groups shown in Table 2. The first group comprises normal directions of magnetization occurring in dykes 2, 3, 4 and in specimen groups 1A and 1B. The second group consists of reversed directions associated with the majority of specimens from dykes 5, 6, 1 and 7. These two groups differ principally in polarity, one being in broad terms a reversal of the other. Groups III and IV consist of reversed directions in the remaining specimens from dykes 5, 6, 1 and 7. These directions differ significantly from those of the first two groups. Comparison of the results of 1974 and 1975 collections shows that they are consistent for dykes 3, 4 and 6 and

Table 1. Mean directions after AF or thermal demagnetization.

Dyke and sample group	Collection	n	N	D	I	α_{95}	k
1A	1974(b,c)	3	13	44.6	74.9	5.8	51.5
1B	1974(b,c)	3	5	354.3	67.1	15.5	25.3
1C	1975(b)	4	21	210.5	-79.9	2.0	254.7
1D	1975(b)	2	8	104.7	-61.7	4.0	193.9
2A	1974(b)	3	9	74.7	76.7	4.4	135.4
2B	1975(b)	4	21	354.7	79.0	3.1	102.8
3A	1974(c)	-	12	356.8	75.3	1.0	1889.0
3B	1975(b)	5	28	0.8	78.5	2.1	167.9
4A	1974(c)	-	13	356.2	80.2	1.5	775.9
4B	1975(b)	4	22	345.2	73.7	1.8	305.4
5A	1974(c)	-	14	29.6	-68.6	3.3	146.0
5B	1975(b)	4	11	173.0	-79.9	4.6	98.3
5C	1975(b)	4	18	300.0	-74.0	5.1	47.1
6A	1974(b)	5	20	211.8	-79.0	1.5	446.2
6B1	1974(b)	5	9	169.9	-79.7	5.7	81.1
6C1	1974(b)	2	9	50.1	-80.6	5.4	91.5
6B2	1975(b)	3	15	170.6	-75.7	2.7	203.3
6C2	1974(b)	3	6	34.9	-77.6	5.9	127.6
7A	1974(c)	-	14	226.6	-80.6	3.3	144.6
7B	1975(b)	4	23	18.7	-74.7	4.8	40.5

Numbers in column 1 refer to dyke numbers as designated in Fig. 1(b), capitals differentiate between significantly different directional groupings, usually associated with different exposures, but sometimes with particular groups of cores from the same or different block samples; column 2 gives the year of collection, while b refers to block samples and c to cores drilled *in situ*; n is the number of block samples and N the number of measurable specimens; computations were performed for N specimens; D is the declination, I the inclination; α_{95} is the radius of 95 per cent circle of confidence, and k is an estimate of the precision parameter (Fisher 1953).

crudely consistent also for dyke 2. There is disagreement, however, between the corresponding directions obtained for dykes 5, 7 and 1, but even these fit the general pattern fairly well.

More detailed considerations show that the directions of specimen groups 5B and 6B represent an almost exact reversal of the directions of specimen groups 1B, 2B and those of dykes 3 and 4. A similar reversal correspondence exists between specimen groups 1A, 2A and 1C, 6A and 7A.

5 Least squares analysis of demagnetization data

To check the validity of the conventional analysis described above a least squares analysis of step demagnetization of some of the specimens was performed, aimed at separating the multi-component phases, especially those with lower coercivity. The method used was developed by Stupavsky & Symons (1978) and is based on the assumption that AF demagnetization processes can be described by mathematical models according to which the direction of each component remains constant as its intensity decays approximately exponentially during demagnetization. The observed remanence vectors are tested for a best fit to five possible models, comprising one, two or three components of different direction, magnitude and stability. The statistical significance of each of the five models can be tested for a minimum of five demagnetizing steps, but eight or more are preferable. The directional goodness of fit is estimated by a parameter referred to as the *mean angular deviation* (MAD) given in degrees of arc. A small MAD indicates a good fit of the directions of the components of NRM to

Table 2. Comparison of basic directional trends.

Directional group	Collection	Dyke and specimen group	n	D	I	α_{95}	k
	1974	1B	5	354.3	67.1	15.5	25.3
	1975	2B	21	354.7	79.0	3.1	102.8
	1974	3A	12	356.8	75.3	1.0	1889.0
I	1975	3B	28	0.8	78.5	2.1	167.9
	1974	4A	13	356.2	80.2	1.5	775.9
	1975	4B	22	345.2	73.7	1.8	305.4
	1974	1A	13	44.6	74.9	5.8	51.5
	1974	2A	9	74.7	76.7	4.4	135.4
*N Mean for 1B, 2B, 3A, 3B, 4A, 4B			6	354.1	75.7	4.1	262.0
N1 Mean for 1A and 2A			2	58.7	76.3	---	---
	1975	5B	11	173.0	-79.9	4.6	98.3
	1974	6B1	9	169.9	-79.7	5.7	81.1
II	1975	6B2	15	170.6	-75.7	2.7	203.3
	1975	1C	21	210.5	-79.9	2.0	254.7
	1974	6A	20	211.8	-79.0	1.5	446.2
	1974	7A	14	226.6	-80.6	3.3	144.6
*R Mean for 5B, 6B1 and 6B2			3	171.1	-78.4	3.6	1150.2
R2 Mean for 1C, 6A and 7A			3	215.9	-79.9	2.6	2178.3
	1975	5A	14	29.6	-68.6	3.3	146.0
III	1974	6C1	9	50.1	-80.6	5.4	91.5
	1974	6C2	6	34.9	-77.6	5.9	127.6
	1975	7B	23	18.7	-74.7	4.8	40.5
*R1 Mean for group III			4	31.2	-75.6	6.7	188.9
IV	1975	1D	8	104.7	-61.7	4.0	193.9
	1975	5C	7	300.0	-74.0	5.1	47.1

*These mean directions do not represent statistical means for the dykes with each dyke treated as a unit and are quoted here to allow a comparison between the principal directional groups; *n* is the number of specimens in the given specimen group or the number of specimen groups used in computing the mean directions; other symbols as in Table 1.

the model being tested. In this analysis it was found convenient to reject values of MAD exceeding 15°.

The analysis was applied to those specimens which, with a few exceptions, had been taken through at least six demagnetizing steps. According to Stupavsky (1980, private communication), a least squares analysis is not suitable for thermal demagnetization data when the decay of remanence cannot be assumed to be exponential. In the present case, however, it was found that the analysis worked quite well also on data obtained by thermal demagnetization.

The results of the least squares analysis are summarized in Table 3. Table 4 lists the mean directions quoted in Table 3 and compares these with the mean directions obtained for the corresponding dykes or groups of specimens using conventional analysis. Although the least squares results have a lower statistical significance than those obtained from conventional procedures, there is a reasonable agreement between the directions obtained by both methods for each dyke or group of specimens investigated except for dyke 1, group 2A and perhaps group 7B. The mean directions obtained by a least squares analysis are very similar for dykes 1, 2 and 4 and groups 3A, 5B reversed and 6B reversed. Least squares analysis has, in fact, singled out successfully the 'characteristic', i.e. supposedly primary components in dyke 1 and in group A of dyke 2, neither of which could be separated by conventional procedure.

A final comparison between the two methods of analysis is made in Table 5. The table lists mean directions with associated statistics and pole positions obtained from the results of conventional analysis for six dykes or groups (nos 1, 2B, 3, 4, 5B and 6B) and for five omitting dyke 1, and also separately for five units using group 3A, instead of all the specimens from dyke 3. Group 1B has been excluded, because it represents only five selected

Table 3. Summary of the least squares analysis.

Dyke and group	D	I	$N(N_T)$	m	Model	Average MAD
1A	358.1	67.1	12	8 or 9	4 or 5	4.15 ± 3.56
1C	177.9	-74.7	6	6, 11 or 12	4 or 5	2.22 ± 1.18
2A	344.7	65.9	4	6 to 9	4	10.84 ± 4.30
2B	337.1	78.7	7	6 or 11	4 or 5	2.60 ± 3.02
3A	354.1	69.5	5(2)	6 to 9	4 or 5	2.64 ± 2.95
3B	20.2	75.3	8(1)	6, 7 or 11	4 or 5	1.81 ± 2.70
4A	349.1	78.1	5(2)	6 to 9	4 or 5	0.87 ± 0.79
4B	343.4	74.6	9(1)	5, 6 or 11	4 or 5	1.18 ± 2.00
5A	7.6	-71.0	8(3)	6 to 8	3 or 4	8.62 ± 9.04
5B	143.6	-66.3	5	6 or 11	4 or 5	3.51 ± 5.33
6B1/A	146.1	-75.5	9	8	2 to 5	3.80 ± 3.41
6B2	166.8	-76.6	6(1)	5, 6 or 11	4 or 5	0.77 ± 0.16
7A	252.2	-64.1	6(4)	6 to 9	4 or 5	5.31 ± 8.37
7B1	71.9	-70.8	7	6 or 11	4 or 5	2.93 ± 2.22
7B2	227.4	-49.5	3(1)	6 or 11	4 or 5	1.84 ± 1.49

N is the number of specimens in each sampling group being analysed, N_T is the number out of N that have been thermally demagnetized; m is the number of demagnetizing steps used; the model numbers quoted are those most commonly fitted and the average value of MAD (mean angular deviation) has been computed for N specimens; group 7B has been subdivided into two sub-groups with distinctly different declinations; other symbols as in Table 1.

Table 4. Comparison of conventional with the least squares analyses.

Dyke or group	Collection year(s)	Conventional analysis N	D	I	α_{95}	k	Least squares analysis N	D	I	α_{95}	k
1A	1974	13	44.6	74.6	5.8	51.8	12	358.1	67.1	14.9	9.5
1C	1975	21	*30.5	79.9	2.0	254.7	6	*357.9	74.7	29.6	6.1
1	1974/75	32	38.3	77.6	2.6	96.2	18	358.0	69.8	12.6	8.5
2A	1974	9	74.7	76.7	4.4	135.4	4	344.7	65.9	7.9	138.1
2B	1975	21	354.7	79.0	3.1	102.8	7	337.1	78.7	8.9	46.5
2	1974/75	--	--	--	--	--	11	341.3	74.1	6.8	46.2
3A	1974	12	356.0	75.3	1.0	1809.0	5	354.1	69.5	7.9	97.0
3B	1975	28	0.8	78.5	2.1	167.9	8	20.2	75.3	7.6	54.2
3	1974/75	36	0.1	77.7	1.7	206.9	13	8.3	73.4	5.6	56.5
4A	1974	13	356.2	80.2	1.5	775.9	5	349.1	78.1	10.9	50.1
4B	1975	22	345.2	73.7	1.8	305.4	9	343.4	74.6	3.5	215.5
4	1974/75	35	348.1	76.2	1.6	238.1	14	345.1	75.8	3.9	104.9
5A	1974	14	29.6	-68.6	3.3	146.0	8	7.6	-71.0	7.7	52.6
5B	1975	11	*353.0	79.9	4.6	98.3	5	*323.6	66.3	19.9	15.7
6B1	1974	9	*349.9	79.7	5.7	81.1	9	*326.1	75.1	9.7	29.2
6B2	1975	15	*350.6	75.7	2.7	203.3	6	*346.8	76.6	4.6	216.2
6B	1974/75	24	*350.4	77.2	2.6	127.0	15	*334.1	76.1	5.8	45.0
7A	1974	14	226.6	-80.6	3.3	144.0	6	252.8	-64.0	15.7	19.1
7B	1975	23	18.7	-74.7	4.8	40.5	7	71.9	-70.8	22.9	7.9
7B	1975	23	18.7	-74.7	4.8	40.5	3	227.5	-49.5	49.5	7.9

For proper comparison the directions marked by an asterisk have been reversed; the directions used in computation of overall mean directions quoted in Table 5 have been underlined; all symbols as in Table 1.

samples. These three results are compared with the corresponding result for six dykes or groups (nos 1, 2, 3A, 4, 5B and 6B) obtained using a least squares analysis. The results of the least squares analysis give closely similar directions and pole positions to those obtained by conventional analysis using five units. The conventional analysis yields, however, a somewhat higher inclination and corresponding higher latitude of the palaeomagnetic pole. Three level

Table 5. Summary of the results.

Dykes or groups	N	D	I	α_{95}	k	ϕ'	λ'	$\delta p, \delta m$
Conventional analysis:								
1, 2B, 3, 4, 5B, 6B	6	0.3	78.4	3.4	382.6	194.5	80.3	6.1, 6.4
2B, 3, 4, 5B, 6B (a)	5	353.1	77.9	1.7	2021.4	210.2	79.3	3.0, 3.2
2B, 3A, 4, 5B, 6B (b)	5	352.5	77.5	1.9	1545.7	210.8	78.6	3.3, 3.6
Least squares analysis:								
1, 2, 3A, 4, 5B, 6B	6	342.7	72.3	4.8	193.5	222.6	69.5	7.6, 8.5
Mean pole (conv. (a))	5	–	–	3.0	649.7	209.8	79.8	– –
Mean pole (conv. (b))	5	–	–	3.5	485.2	210.1	78.9	– –
Mean pole (1. sq.)	6	–	–	7.9	72.8	222.7	69.7	– –

N is the number of dykes or specimen groups used in the computation; ϕ' and λ' are the longitude and latitude of the palaeomagnetic poles in the northern hemisphere; $\delta p, \delta m$ are the semi-axes of the oval of confidence of the pole; other symbols as in Table 1.

statistics (Watson & Irving 1957), applied separately to the two sets of data revealed a significant between site variation in direction for the five unit data set obtained by conventional analysis when group 3B was included, but there was no such significant variation when 3B data were omitted. ($F = 1.839$ was then exceeded by $F_{8,178}$ at the 5 per cent probability level.) Similarly, there was no significant between site variation in the directions obtained by a least squares analysis ($F = 0.896$ was exceeded by $F_{154,10}$ at the 5 per cent probability level). The reason for including the computation for the conventional analysis data for five units from which group 3B data had been excluded is related to the results of these significance tests, but obviously the corresponding mean directions or pole positions differ very little. Table 5 shows that the pole positions obtained from the mean directions are in each case quite similar to the corresponding mean pole positions quoted at the bottom of the table, confirming the low scatter in directions.

6 Discussion

The following conclusions can be drawn from the above-presented analysis of the data: several stable components of NRM have been identified (Table 2) and among these two predominant ones occur, one normally directed in specimen groups 1B and 2B and in dykes 3 and 4 and the other a reversed one in specimen groups 5B and 6B. As shown in Table 5 the directions of these two components were used in computing the final mean direction and mean pole position for the dykes. The other stable components have different directions from the two predominant ones and some differ significantly (e.g. 1D and 5C). Stable normal and reversed components occur only in dyke 1 (e.g. 1A and 1C, Table 2). The occurrence of stable NRM components of variable polarity suggests that this may be a record of changing direction of past geomagnetic field over a relatively short time interval covering the period of field reversal.

Table 2 also gives the mean directions computed for each of the five significant directional groupings. These mean directions are plotted on an equal area net in Fig. 6 which shows that the directions designated N, R1 and R lie on a great circle, while those designated N1 and R2 lie off this circle. It would appear that direction R represents a reversal of direction N and R1 corresponds to a spot record of geomagnetic field direction during its reversal.

There are two alternative explanations of the directions N1 and R2. One suggestion is that these directions represent fluctuations in geomagnetic field direction prior to or during

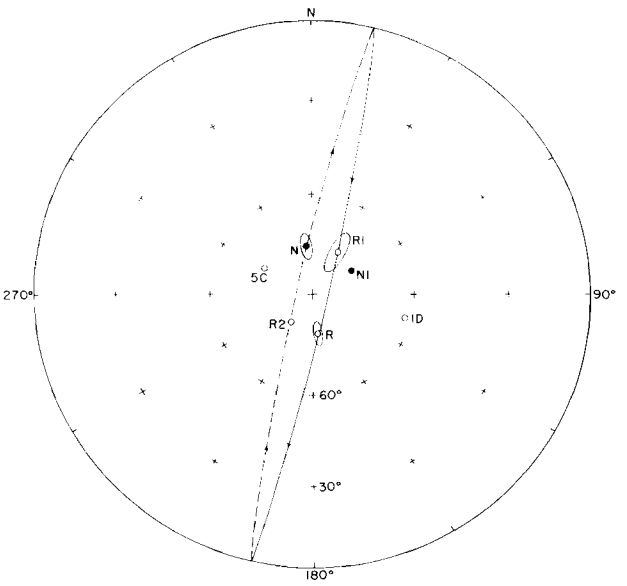


Figure 6. Equal area plot of basic directional trends listed in Table 2. N, mean for specimen groups 1B, 2B, 3A, 3B, 4A and 4B; N1, mean for specimen groups 1A and 2A; R, mean for specimen groups 5B, 6B1 and 6B2; R1, mean for specimen groups 5A, 6C1, 6C2 and 7B and finally R2, mean for specimen groups 1C, 6A and 7A. Full symbols: lower hemisphere; open symbols: upper hemisphere. α_{95} circles shown where possible.

a reversal such as have been observed in recent rocks (Fuller & Williams 1979) and this interpretation would also explain the directions of group IV. It is of interest to note that the observed palaeomagnetic reversal corresponds well with the previously established reversal at the boundary of Aptian and Albian indicated among others on the geomagnetic polarity scale by the M0 oceanic anomaly (Lowrie, Channell & Alvarez 1980). The second explanation takes note of the fact that R2 appears to represent a reversal of N1. It is thus possible to combine the reversed directions of the three units of group R2 with those of N1 to obtain a significant mean direction, suggesting a similar secondary influence during the history of the units considered. This direction with $D = 46.7°$, $I = 78.7°$, $\alpha_{95} = 4.1°$, $k = 347.1$ compares favourably with that obtained by Sandal & Halvorsen (1973) from basalt lavas of the late Mesozoic Seidfell volcano with $D = 48°$, $I = 81°$, $\alpha_{95} = 6.9°$, $k = 44.7$. The latter has been attributed to post-Cretaceous remagnetization, perhaps as a result of the West Spitsbergen orogeny, and it seems possible that the same cause can be invoked in the present case.

Summarizing the main points of this discussion, several explanations of the observed data can be proposed. If it is assumed that the dykes were emplaced during a field reversal, it seems possible that all the 'characteristic' NRM directions are primary. Another plausible explanation suggests that all the dykes were subjected to post-formational remagnetization associated with regional hydrothermal alteration in the late Mesozoic, and this may also have occurred during a geomagnetic field reversal. Several other mechanisms are possible. For instance the normal and reversed directions (N and R in Fig. 6) might be primary and the other directions (N1, R1, R2 and those of specimen groups 1D and 5C) secondary, imposed after emplacement of the dykes in the late Mesozoic or during the Tertiary orogeny. As mentioned in Section 3, the degree of hydrothermal alteration is virtually the same for all dykes, irrespective of their position within the Tertiary orogenic belt. This suggests a strong contribution by Mesozoic metasomatism. Nevertheless, the presence of an unaltered fraction

of class 1 or 2 TM suggests that some of the original thermoremanence (TRM) should have been retained. Since dykes 3 and 4 contain finer (1–20 μm) magnetic grains, their NRM is, in fact, quite stable. Moreover, thermomagnetic studies conducted by Jeleńska *et al.* (1978/79) showed that the magnetic minerals in these two dykes were less oxidized than those in the other dykes. It is thus reasonable to conclude that the characteristic NRM direction of these two dykes is primary. Consequently, similar directions in other dykes (e.g. 2B) as well as reversals resulting in analogous directions (specimen groups 5B and 6B) are also accepted as primary. The question of the origin of the other directions remains open, since there is insufficient evidence to show that they are either primary or secondary.

It is of interest to mention at this juncture the generally lower stability secondary NRM components identified by a least squares analysis in dykes 5 and 7. In specimen group 5A a component with $D = 147.3°$, $I = 70.4°$, $\alpha_{95} = 4.2°$ was clearly separated. Apart from the directions quoted in Table 4, the two most prominent lower stability components in 7A had directions characterized by $D = 76.2°$, $I = 87.7°$, $\alpha_{95} = 6.8°$ and $D = 275.0°$, $I = 81.0°$, $\alpha_{95} = 12.5°$. In 7B a component with $D = 59.3°$, $I = 60.9°$, $\alpha_{95} = 22.1$ was identified. All these components clearly do not involve viscous remanence imposed by the recent geomagnetic field, but probably represent effects of Tertiary orogeny. In addition several other rather unstable components were identified whose origin appears to be related to imperfections in the demagnetization procedure (Wilson & Lomax 1972; Stephenson 1976).

The above discussion supports the conclusion that the directions isolated by conventional procedure from dykes or specimen groups 2B, 3, 4, 5B and 6B and those obtained by a least squares analysis from dykes or groups 1, 2, 3A, 4, 5B and 6B are representative for the direction of the late Mesozoic geomagnetic field in Spitsbergen. The mean pole positions given in Table 5 are similar and compare favourably with the pole position recently obtained from

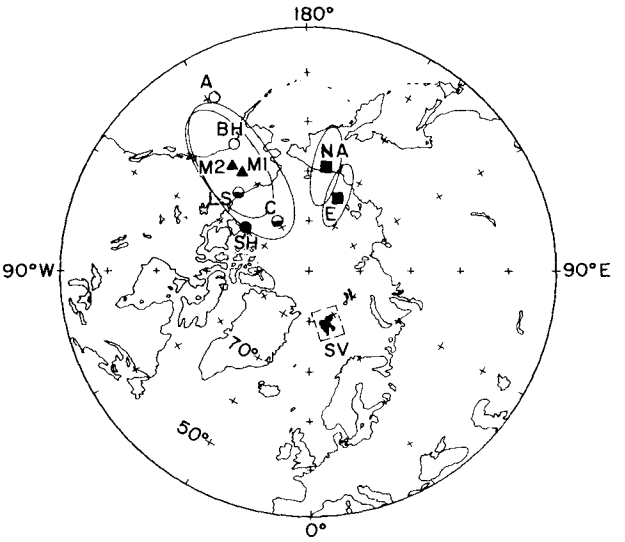


Figure 7. Summary of palaeomagnetic results. Mean pole positions: Spitsbergen dolerites and lavas: normal polarity, full circles; reversed polarity, open circles; mixed polarity, half-full circles. This study: conventional analysis, C; least squares analysis, LS. Other studies: A, Agardhdalen sills (Vincenz *et al.* 1981); SH, Sandal & Halvorsen (1973); BH, Briseid & Halvorsen (1974). Means of four poles: triangles, M1, using pole C and M2, using pole LS. Full squares: NA, mean Cretaceous pole for North America for the time interval 83 to 159 Myr BP; E, mean Cretaceous pole for Eurasia for approximately the same time interval (Irving 1980); SV, Svalbard Archipelago. Ninety-five per cent circles of confidence for M1, M2, NA and E are also shown.

dolerite sills of Agardhdalen (Vincenz *et al.* 1981) (longitude 208.2°, latitude 50.5° N) as well as with pole positions obtained by Sandal & Halvorsen (1973) from the Mesozoic Sörlifjell lavas (longitude 235°, latitude 75° N) and by Briseid & Halvorsen (1974) from the late Jurassic dolerite sill of Lomfjord in North-east Spitsbergen (longitude 210°, latitude 61° N). A comparison of these pole positions is made in Fig. 7 which also shows the mean pole position derived from the four poles as well as the relevant circle of 95 per cent confidence. Two mean pole positions are given, one based on the results of a conventional analysis (b in Table 5) at 213.3° and 66.7° N, $\alpha_{95} = 15.8°$ and the other on the results of a least squares analysis at 215.6° and 64.4° N, $\alpha_{95} = 13.5°$. Clearly the difference between the two positions is not significant and each can be accepted as the late Mesozoic palaeomagnetic pole for Svalbard.

7 Conclusions

The positions of late Mesozoic palaeomagnetic poles based on the results from Spitsbergen dolerites are compared in Fig. 7 with those of the Cretaceous poles for North America and Eurasia. The latter are based on the most recent data available (Irving 1980). The North American pole is a mean of eight poles ranging in age from 83 to 159 Myr BP while the Eurasian pole is a mean of fifteen poles spanning the interval from 87 to 159 Myr BP. The Mesozoic poles for Spitsbergen are distinct from both the North American mean Cretaceous pole as well as from the Eurasian pole. Although the data are still too scarce to allow any firm conclusions of palaeotectonic nature, they support the suggestion (Harland 1969; Sandal & Halvorsen 1973) according to which the opening of the Neo-Arctic basin was a Tertiary event. This is suggested by the fact that the Cretaceous North American pole lies west of the Spitsbergen pole and hence North America and Spitsbergen and Eurasia must have been pulled apart in post-Cretaceous times.

The clear indication of field reversals recorded correlated with K-Ar ages obtained from dyke 6, places the time of reversal within a 10 Myr period centred on 110 Myr BP and coinciding with the well-established early Cretaceous reversal event (Lowrie *et al.* 1980).

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